



HRISHITAA NALAWADE

Machine Learning Engineer | AI Specialist | Python Developer

CONTACT

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EDUCATION

MIT ADT University 2023 – 2027

B.Tech Computer Science
Engineering (AI in Analytics) | CGPA:
8.49/10

SKILLS

- AI / GenAI: LLM Integration · RAG (Retrieval-Augmented Generation) · Agentic AI · Multi-Agent Orchestration · NLP · Transformers · HuggingFace · Prompt Engineering · Intent Classification · scikit-learn · Transfer Learning · Deep Learning
- Backend & APIs: FastAPI · Flask · REST APIs · API Design · Authentication · Caching · System Design · Scalability · Microservices
- Languages: Python · Java · SQL · HTML · CSS
- Databases: PostgreSQL · MySQL · Database Design
- Cloud & DevOps: AWS (Cloud Foundations Certified) · Pantheon · Git · GitHub · Postman
- Data & BI: Tableau · Power BI · Excel · Jupyter Notebook
- Responsible AI: Human-in-the-Loop (HITL) · Agent Observability · SLA Management · Governance

CERTIFICATIONS

- Programming Essentials in Python – Cisco Networking Academy
- Google AI Essentials – Google
- Python for Data Science and Artificial Intelligence – IBM
- Introduction to Tableau – Tableau Learning Partner
- SQL (Basic) Certificate – HackerRank

SUMMARY

AI Engineering student (B.Tech CSE – AI in Analytics, CGPA 8.49) with production experience building LLM-powered applications, agentic systems, and RAG pipelines. Architected and deployed retrieval-augmented generation chatbots and LLM orchestration layers at Acuta Digital Inc. Skilled in multi-agent system design, FastAPI, NLP, and intelligent workflow automation.

WORK EXPERIENCE

Acuta Digital Inc. | Jun 2025 – Jul 2025 | Enterprise Production Environment
AI Integration Intern | Mentorship | Real-world Project Experience

- Developed and deployed an AI-powered conversational chatbot module integrated within a Drupal enterprise CMS on the Pantheon platform, writing clean, efficient, and maintainable Python code while collaborating with cross-functional teams across a global enterprise environment.
- Engineered end-to-end LLM (Large Language Model) integration architecture encompassing backend logic, API layers, and RESTful endpoints; applied NLP and Retrieval-Augmented Generation (RAG) pipelines using Transformers (HuggingFace) to process and respond to natural language user queries with high accuracy.
- Designed and implemented scalable Flask-based backend systems for content retrieval and query processing; optimised algorithms for efficiency and responsiveness, ensuring high performance and quality of the application under varying production loads.
- Collaborated with product managers, designers, and developers using Agile methodologies; participated in code reviews, provided constructive feedback, and iteratively improved chatbot response accuracy through data-driven optimisation and knowledge-sharing.
- Gained hands-on expertise in enterprise deployment workflows, API design patterns, and IT service frameworks, applying continuous improvement practices to enhance both system reliability and user engagement metrics.

PROJECTS

AI-Powered Agent Monitoring System | 2026 | Enterprise-Scale Application
Technologies: Python | FastAPI | PostgreSQL | REST APIs | System Design | Real-time Performance Monitoring | Scalability

- Architected a production-ready, layered REST API (Router → Schema → Service → ORM → DB) to monitor distributed AI agents — tracking real-time health status, uptime, and performance indicators across agent ecosystems; built on FastAPI with SQLAlchemy ORM and PostgreSQL for persistent agent state management.
- Integrated OpenRouter LLM via async HTTPX calls to auto-generate intelligent agent health summaries at the `/agents-summary/` endpoint; implemented response caching to reduce redundant LLM calls and optimize both cost and latency at scale.
- Enforced enterprise-grade security and reliability using bearer token authentication, strict Pydantic schema validation at all system boundaries, and SlowAPI-based rate limiting to protect system resources from abuse.
- Designed full agent lifecycle management via RESTful endpoints — agent registration, real-time status updates, retrieval, and deletion — with filtering and pagination support; async-first design ensures non-blocking performance under concurrent load.

AI-Based Ticketing System (Conversational Chatbot) | Python · FastAPI · PostgreSQL · OpenRouter LLM · Bearer Token Auth · HITL Workflow

Technologies: Python | FastAPI | PostgreSQL | SQLAlchemy | ORM | OpenRouter LLM (GPT-4o-mini)
NLP Bearer Token Auth | REST APIs | HITL Workflow | SLA Management

- Engineered an LLM-powered support ticketing backend using OpenRouter (GPT-4o-mini) for automated ticket classification by category (Billing / Technical / General) and urgency (High / Medium / Low), plus AI-generated draft replies — with graceful fallback logic when LLM responses are missing or malformed.
- Implemented a Human-in-the-Loop (HITL) review workflow where all new tickets enter a `needs_review` state, enabling human reviewers to approve, correct category/urgency, or edit AI draft replies before final approval or closure — ensuring responsible AI output validation before action.
- Built SLA deadline tracking tied to urgency tier (High: 1hr / Medium: 4hr / Low: 24hr) using PostgreSQL timestamp logic, with automated SLA breach detection, escalation log creation for high-priority tickets, and real-time breach badges surfaced in the frontend dashboard.
- Designed a comprehensive REST API surface — create, list, filter by category/urgency/status/SLA state, sort, full-text search across ticket text and draft replies, stats endpoint, and a PATCH review endpoint — all secured with bearer token auth and accessible via a lightweight browser-based HTML/CSS/JS frontend.